

BrightLearn Data Analytics

Advance SQL : Null Functions

Q1 : Show all employees salary with their salary. If salary is NULL, display 0.

```
SELECT employee_id, name,  
       IFNULL (Salary, 0) AS salary_with_default  
FROM workspace.brightlearn.employees;
```

employee_id	name	salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2 Show employee names with their department. If department is NULL, show "Not Assigned".

```
SELECT employee_id, name,  
       IFNULL (department, 'Not Assigned') AS department_name  
FROM workspace.brightlearn.employees;
```

Q3 Find orders with NULL customer_id using ISNULL()

```
SELECT order_id, customer_id  
FROM workspace.brightlearn.null_orders  
WHERE ISNULL (customer_id);
```

order_id	customer_id
103	NULL

Q4 Show all orders if delivery-date is NULL Show 'Pending'

```
SELECT order_id, customer_id,
       IFNULL(CAST(delivery_date AS STRING), 'Pending') AS delivery_status
FROM workspace.brightlearn.nullorder;
```

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Q5 Show all students and their grades Replace NULL with 0.

```
SELECT student_id, name,
       IFNULL(grade, 0) AS final_grade
FROM workspace.brightlearn.students
WHERE grade IS NULL;
```

student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6 Count students who haven't been graded.

```
SELECT COUNT(*) AS not_graded_count
FROM workspace.brightlearn.students
WHERE grade IS NULL;
```

not_graded_count
1

Q7 Show product name, price, and final price after discount. Assume 0 if discount is NULL.

```
SELECT product_id, name,  
       price - IFNULL(discount, 0) AS final_price  
FROM workspace.brightlearn.products;
```

Product_id	name	final_price
501	keyboard	25
502	Mouse	10
503	Monitor	100

Q8. Count how many customers have no email.

```
SELECT COUNT(*) AS missing_email_count  
FROM workspace.brightlearn.customers;  
WHERE email IS NULL;
```

missing_email_count
2

Q9 Show all customers with email. If NULL, display 'No Email'.

```
SELECT customer_id, name,  
       IFNULL(email, 'No email') AS email_display  
FROM workspace.brightlearn.customers;
```


customer_id	name	email_display
1	Linda	No Email
2	Jo Seph	Joseph @ mail . com
3	Nia	No Email

Q 10 Show payment details with method replaced by "Unknown" if null.

```
SELECT payment_id
       IFNULL (method, 'Unknown') AS method_display, Status
FROM workspace.brightlearn.Payments;
```

Payment-id	method_display	Status
301	Credit	NULL
302	Pay Pal	Success
303	Unknown	Failed

Q 11 Show item and their quantity (0 if Null).

```
SELECT item_id, item_name,
       IFNULL (quantity, 0) AS quantity_checked
FROM workspace.brightlearn.Inventory;
```

item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Erasor	0

Q 12 show emp id
or commission

```
SELECT emp_id,
       (CASE WHEN commission IS NULL THEN 0 ELSE commission END) AS first_available_reward
FROM workspace.brightlearn.EmployeeExtra;
```

emp_id	first_available_reward
1	300
2	100
3	NULL

Q 13 Count classes that don't have a assigned.

```
SELECT COUNT(*) AS no_room_count
FROM workspace.brightlearn.Classes
WHERE room IS NULL;
```

no_room_count
2

Q 14 Show attendance records with status. Replace NULL with 'Not marked'

```
SELECT student_id, date,
       IFNULL(attendance_status, 'Not Marked') AS attendance_status
FROM workspace.brightlearn.Attendance;
```

student_id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Q 15 Show account id, account-type (or 'unknown') and balance (or 0).

```
SELECT account_id,  
       IFNULL ( id Account_type, 'Unknown') AS type_display,  
       IFNULL ( balance, 0 ) AS balance_checked  
FROM workspace.brightlearn.Bank_Accounts;
```

account_id	type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Q 16 Show all projects with a start date. If start-date is NULL, display 'TBD'.

```
SELECT project_id, title,  
       IFNULL (CAST (start_date AS STRING), 'TBD')  
AS start_display  
FROM workspace.brightlearn.Projects;
```

Project_id	title	start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Q 17 Display reviews showing comment (or 'No comment') and rating (or)

```
SELECT review_id, product_id,
       IFNULL (comment, 'No comment') AS comment_display,
       IFNULL (rating, 0) AS rating_display
FROM workspace.brightstern_suppliers_reviews;
```

review_id	product_id	comment_display	rating_display
1	S01	Great Product	4
2	S02	No comment	0
3	S03	Works fine	3

Q18 Show Supplier contact number using COALESCE (phone, alt_phone, 'No Contact').

```
SELECT Supplier_id,
       name,
       COALESCE (phone, alt_phone, 'No Contact') AS contact_number
FROM Suppliers;
```

Supplier_id	name	Contact-number
1	Global Goods	123456789
2	Best Supplies	987654321
3	Value Source	No contact

Q 19 Show all users and their preferences. Replace Nulls : Theme → 'light',
Language → 'English', Timezone → 'UTC'.

```
SELECT user_id
       IFNULL (theme, 'light') AS theme_set,
       IFNULL (Language, 'English') AS language_set,
       IFNULL (timezone, 'UTC') AS timezone_set
FROM User_Settings;
```

User_id	theme_set	Language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC +1
3	Light	English	UTC

Q 20 Show maintenance log. Issue → 'Unknown Issue', technician → 'Not Assigned'.

```
SELECT record_id,
       Machine_id
       IFNULL (issue 'Unknown Issue') AS issue_Log
       IFNULL (technician 'Not Assigned') AS technician_name
FROM Maintenance;
```

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex